EarthTrend Explorer

Scientific Project Description & Technical Evaluation Dossier

NASA Space Apps Challenge: Be An Earth System Trend Detective!

Team: Machine Without Learning

GitHub Repository: https://github.com/ammarsami255/Be-An-Earth-System-Trend-Detective

Live Web Application: https://earthtrend-explorer.streamlit.app/

Evaluation Status: 71/71 Tests Passing (100% Pass Rate) · Clean Verification

---

1. Executive Summary & Challenge Alignment

EarthTrend Explorer is a scientifically grounded climate intelligence platform and empirical statistical laboratory. Utilizing 43 years (1981–2023, 516 contiguous monthly observations per variable, validated for completeness and missing-value handling) of authentic NASA Earth observation data (NASA POWER / MERRA-2) across three representative Egyptian climate zones—Greater Cairo (urban valley), Alexandria (Mediterranean coast), and Aswan (hyper-arid desert)—the platform investigates long-term environmental shifts, abrupt hydrologic changes, and predictive climate anomalies.

Addressing the NASA Challenge: The platform rigorously answers the four foundational questions posed by the challenge:

- **1. What is changing?** Mean temperatures (`T2M`), absolute monthly extreme peak hours (`T2M_MAX` , `T2M_MIN`), physical precipitation totals (`PRECTOTCORR` in mm/month and mm/year), multi-scalar drought indices (SPI-3/6/12), and compound hot-dry extremes.
- **2. Where is it changing?** Evaluated across three distinct Egyptian regional regimes mapped to native NASA MERRA-2 0.5° × 0.625° reanalysis grid cells.
- **3. How much is it changing?** Quantified using the robust non-parametric Theil-Sen median slope estimator with Gilbert (1987) 95% confidence intervals, decadal rates, and change-point magnitudes.
- **4. Is it statistically significant?** Evaluated via the non-parametric Mann-Kendall test, Hamed-Rao (1998) serial autocorrelation corrections, Hirsch-Slack Seasonal Mann-Kendall tests, Benjamini-Hochberg FDR ($q < 0.05$) and Holm-Bonferroni FWER ($\alpha < 0.05$) multiple testing controls, and the Pettitt (1979) change-point test.

Why It Matters: Regional climate adaptation in vulnerable, water-stressed arid environments requires transparent, physically guarded statistical tools rather than unverified black-box claims. By identifying steady warming (+0.26 to +0.43 °C/decade), an abrupt 2015 rainfall step-change (+181% to +266%), and surging compound hot-dry extremes, EarthTrend Explorer converts authentic NASA observations into reliable, localized climate intelligence for water resource planning and regional resilience.

2. Project Details: Architecture, Features, & Status

Operational Architecture & Methodology

EarthTrend Explorer functions as a unified pipeline structured across four analytical modules:

1. **Data Ingestion & Physical Conversion:** Ingests monthly air temperature (`T2M`), absolute monthly extreme peak hours (`T2M_MAX` , `T2M_MIN`), and bias-corrected precipitation flux (`PRECTOTCORR`), converting daily rainfall rates (mm/day) into physical monthly accumulation (mm/month) with leap-year adjustments.
2. **Non-Parametric Trend Engine:** Assesses long-term trends without assuming normality. Accounts for rank autocorrelation using the Hamed-Rao modified variance $\text{Var}^*(S)$, uncovering that lag-12 autocorrelation in Alexandria ($\rho_{12} = -0.301$) deflates variance and raises Z^* from +5.50 to +6.23.
3. **Anomaly & Regime Shift Detection:** Computes WMO 1981–2010 monthly climatological normal departures (μ_m, σ_m), tracks calendar-continuous persistent heat/cold spells (≥ 3 months, $|Z| \ge 1.0\sigma$), applies Pettitt change-point detection to identify hydrologic regime breaks, and fits 2-parameter Gamma distributions for Standardized Precipitation Index (SPI) drought monitoring.
4. **Predictive Anomaly Forecasting:** Implements three candidate time-series models (Seasonal Naive baseline, Holt-Winters Additive Exponential Smoothing, and SARIMAX). Automatically selects the tournament winner via expanding-window chronological cross-validation across 5 historical origins (2018–2022) using Mean Absolute Scaled Error (MASE). Computes 80% and 95% prediction intervals and evaluates forward projections against the WMO 1981–2010 baseline.

Feature Status, Limitations, & Future Roadmap

Feature Component	Status	Implementation Details	Scientific Consideration / Limitation
Non-Parametric Trends	**Working Feature**	Mann-Kendall, Theil-Sen 95% CIs, Hamed-Rao autocorrelation, Seasonal MK, FDR/FWER across 12 months.	Statistical trends document observational changes; they do not single-handedly establish causal attribution.
Physical Rainfall Totals	**Working Feature**	Converted daily rate (mm/day) to physical monthly/annual accumulated totals with leap-year accuracy.	Reanalysis grid cells represent area averages, which smooth out local convective cloudbursts.
Pettitt Change-Point Test	**Working Feature**	Detected abrupt step-change in rainfall between 2014 and 2015 in Cairo ($p=0.0067$) and Alexandria ($p=0.0060$).	Detects *when* a statistical break occurred, not *why*. Requires ground-truth validation against EMA gauges.
SPI Drought & Aridity Guardrails	**Working Feature**	Multi-scale SPI (3, 6, 12) via Gamma MLE. Flags SPI as low-reliability when baseline zero frequency $q \ge 0.50$.	In hyper-arid regions (Aswan, zero rain in 86% of months), SPI is mathematically bounded above -1.0σ ($\Phi^{-1}(q)$).

Feature Component	Status	Implementation Details	Scientific Consideration / Limitation
Forecasting Tournament	**Working Feature**	5-origin expanding window backtesting (2018–2022); evaluated by MAE, RMSE, and MASE. Winner: Holt-Winters ETS.	Statistical extrapolations are probabilistic projections with uncertainty, not deterministic future guarantees.
2024 Reality Check	**Working Feature**	Forecasts made from 1981–2023 are scored against real 2024 NASA observations strictly held out from training.	Tests out-of-sample skill and evaluates binomial calibration of 80% and 95% prediction intervals.
GPM IMERG Cross-Validation	**Future Roadmap**	Cross-validate the 2015 MERRA-2 rainfall jump against NASA GPM IMERG satellite constellation observations.	Will distinguish physical climatic shifts from potential reanalysis observing-system sensor transitions.
Spatial NetCDF Grid Ingestion	**Future Roadmap**	Direct ingestion of spatial bounding box NetCDF cubes across Egypt via xarray.	Current version uses verified NASA POWER point extractions at representative coordinates.

Technology & Software Stack

- **Programming Language:** Python 3.10+
- **Data Engineering & Science:** Pandas 2.0+, NumPy 1.24+
- **Statistical & Forecasting Engines:** SciPy 1.11+ (`scipy.stats`), Statsmodels 0.14+ (`statsmodels.tsa`), PyMannKendall 1.4.3
- **Interactive Visualizations:** Plotly 5.18+ (`graph_objects`, `subplots`) tuned to a high-contrast NASA space dark theme
- **User Interface:** Streamlit 1.30+ with a glass top navbar and 9 specialized analytical pages
- **Quality Assurance:** Pytest (71 automated unit, integration, and headless Streamlit AppTest tests passing at 100%)

3. NASA Earth Science Data & Resources Used

NASA Resource Name	Direct Full Link	Role & Utilization in Project
NASA POWER Project (Prediction of Worldwide Energy Resources)	https://power.larc.nasa.gov/	Primary data source for 43-year continuous meteorological time series (1981–2023) across Cairo, Alexandria, and Aswan.
NASA POWER Monthly API Service	https://power.larc.nasa.gov/docs/services/api/temporal/monthly/	Programmatic point API used for automated data ingestion, header

NASA Resource Name	Direct Full Link	Role & Utilization in Project
		metadata extraction, and live coordinate queries.
NASA GMAO MERRA-2 (Atmospheric Reanalysis, Version 2)	https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/	Underlying physical assimilation model providing gridded 0.5° × 0.625° observations: `T2M`, `T2M_MAX`, `T2M_MIN`, and `PRECTOTCORR`.
NASA Earthdata (EOSDIS)	https://www.earthdata.nasa.gov/	Data governance reference and authentication provider for NASA Earth Science data holdings.
NASA Giovanni Portal	https://giovanni.gsfc.nasa.gov/	Supported data format in `src/data_loader.py` for cross-verifying time-series exports and regional bounding boxes.

4. Space Agency Partner, International Standards, & Third-Party References

Resource / Standard	Direct Full Link	Citation & Implementation Credit
WMO Climate Normals Guidelines (WMO-No. 1203 & WMO-No. 49)	https://library.wmo.int/records/item/55797-wmo-guidelines-on-the-calculation-of-climate-normals	World Meteorological Organization standard 1981–2010 30-year climatological normal baseline methodology and ≥10-year provisional normal rule.
WMO SPI User Guide (WMO-No. 1090)	https://library.wmo.int/records/item/39719-standardized-precipitation-index-user-guide	Standard international classification brackets and drought threshold color conventions (Extremely Wet to Extremely Dry).
pyMannKendall Library (Hussain & Mahmud, 2019)	https://github.com/mmhs013/pyMannKendall	Open-source Python library used for baseline non-parametric trend verification and rank correlation routines.
Statsmodels Econometrics Library	https://www.statsmodels.org/	State-space time-series framework implementing Holt-Winters Exponential Smoothing and SARIMAX model estimation.
Academic Literature Citations	SOURCES_AND_REFERENCES.md	Mann (1945), Sen (1968), Pettitt (1979), Hamed & Rao (1998), Hyndman & Koehler (2006, MASE), Box & Jenkins (2015), Pyper & Peterman (1998, N_{eff}).

5. Disclosure of Super Intelligence (SI) & Artificial Intelligence Tools

In compliance with the NASA Space Apps Challenge Super Intelligence Policy:

- **How SI was Utilized:** Large Language Model (LLM) coding assistants were used collaboratively for code scaffolding, automated unit test generation in `pytest`, and formatting technical documentation and Markdown dossiers.
- **Strictly Human & Deterministic Elements:**
  - **Data Authenticity:** Zero data was generated, synthesized, or simulated by AI. All inputs are 100% authentic NASA POWER / MERRA-2 observations.
  - **Mathematical Rigor:** All trend statistics (S, Z^*, Theil-Sen slope), change-point ranks (K), drought indices (SPI), and forecast predictions are executed deterministically by Python numerical libraries (SciPy, Statsmodels, NumPy).
  - **Epistemological Candor:** The project strictly refrains from using generative AI to produce unverified causal attributions or fabricate environmental phenomena.
- **Branding Compliance:** No official NASA logos, insignias, or mission branding were generated, modified, or placed in project media.

> **Project Verification & Review Links:**
> • **Public GitHub Repository:** https://github.com/ammarsami255/Be-An-Earth-System-Trend-Detective
> • **Live Interactive App:** https://earthtrend-explorer.streamlit.app/ (Deployed & Live on Streamlit Community Cloud)
> • **Master Reproducible Audit Script:** `python scripts/reproducible_scientific_audit.py`